

Week 40: More on continuous-time Markov Chains

Repetition from last week

Definition 0.1. (*Continuous-time Markov Chain*) A stochastic process $\{X(t) : t \geq 0\}$ with discrete state space $\mathcal{S}$ that satisfies the Markov property and has stationary transition probabilities.

Recall:

- Markov property:

$$P(X(s+t) = j \mid X(s) = i, X(u) = x(u), 0 \leq u < s) = P(X(t+s) = j \mid X(s) = i).$$

- Stationary transition probabilities:

$$P(X(s+t) = j \mid X(s) = i) = P(X(t) = j \mid X(0) = i).$$

- Transition probability function $P_{i,j}(t) = P(X(t) = j \mid X(0) = i)$

Repetition from last week

Theorem 0.2. *In a birth-and-death process with birth rates λ_i and death rates μ_i , the following holds:*

- *Independent sojourn times.*
- $S_i \sim \text{Exp}(\lambda_i + \mu_i)$ for $i \geq 0$.
- *Jump probabilities in state i*

$$P(i \rightarrow i - 1) = \frac{\mu_i}{\lambda_i + \mu_i} \quad \text{and} \quad P(i \rightarrow i + 1) = \frac{\lambda_i}{\lambda_i + \mu_i}.$$

This week

Theorem 1.3. (*Forward Kolmogorov differential equations*) For a birth-and-death process on $\mathcal{S} = \{0, 1, 2, \dots\}$, the $P_{i,j}(t)$ satisfy

$$P'_{i,0}(t) = -\lambda_0 P_{i,0}(t) + \mu_1 P_{i,1}(t), \quad t \geq 0$$

$$P'_{i,j}(t) = \lambda_{j-1} P_{i,j-1}(t) - (\lambda_j + \mu_j) P_{i,j}(t) + \mu_{j+1} P_{i,j+1}(t), \quad t \geq 0, \quad j \geq 1,$$

with initial conditions $P_{i,j}(0) = \delta_{i,j}$.

Trick: Solve using integrating factor. For ODE/IVP

$$\frac{dy}{dx} + P(x)y = Q(x)$$

we multiply each side by $\mu(x) = \exp(\int P(x) dx)$ and use the product rule.

Problem

A machine works (1) or is broken (0). Time to breakdown is $\text{Exp}(\mu)$, and time to repair is $\text{Exp}(\lambda)$, and all sojourn times are independent. The machine works at time $t = 0$. Calculate the probability that the machine works at time $t = 10$.

Problem

A pure birth process starting from $X(0) = 0$ has birth parameters

$$\lambda_0 = 1, \quad \lambda_1 = 3, \quad \lambda_2 = 2, \quad \lambda_3 = 5.$$

Let W_1 , W_2 and W_3 be the stochastic times it takes the process to reach states 1, 2 and 3.

1. Determine the transition probability functions

$$P_n(t) = P(X(t) = n | X(0) = 0), \quad n = 0, 1, 2, 3.$$

2. Write W_3 as a sum of sojourn times and thereby deduce that the mean time is $\mathbb{E}[W_3] = 11/6$.
3. Determine the mean of $W_1 + W_2 + W_3$.
4. Determine the variance of W_3 .

Theorem 1.4. *For a birth-and-death process without absorbing states,*

$$\pi_j = \lim_{t \rightarrow \infty} P_{i,j}(t), \quad j = 0, 1, \dots,$$

exists and are not dependent on i .

Remarks:

- *If $\sum_{j=0}^{\infty} \pi_j = 1$, the $\boldsymbol{\pi} = (\pi_0, \pi_1, \dots)^\top$ is called a limiting distribution.*
- *If the limiting distribution exists, it is also a stationary distribution*

$$\pi_j = \sum_{i=0}^{\infty} \pi_i P_{i,j}(t), \quad j = 0, 1, \dots, \quad t \geq 0.$$

Towards some practical formulas

- Kolmogorov forward equations

$$P'_{i,0}(t) = -\lambda_0 P_{i,0}(t) + \mu_1 P_{i,1}(t), \quad t \geq 0$$

$$P'_{i,j}(t) = \lambda_{j-1} P_{i,j-1}(t) - (\lambda_j + \mu_j) P_{i,j}(t) + \mu_{j+1} P_{i,j+1}(t), \quad t \geq 0, \quad j \geq 1,$$

- Take the limit $t \rightarrow \infty$ and use that $P'_{i,j}(t) \rightarrow 0$

$$0 = -\lambda_0 \pi_0 + \mu_1 \pi_1,$$

$$0 = \lambda_{j-1} \pi_{j-1} - (\lambda_j + \mu_j) \pi_j + \mu_{j+1} \pi_{j+1}, \quad j \geq 1.$$

- Reorganize to get

$$\lambda_0 \pi_0 = \mu_1 \pi_1,$$

$$(\lambda_j + \mu_j) \pi_j = \lambda_{j-1} \pi_{j-1} + \mu_{j+1} \pi_{j+1}, \quad j \geq 1.$$

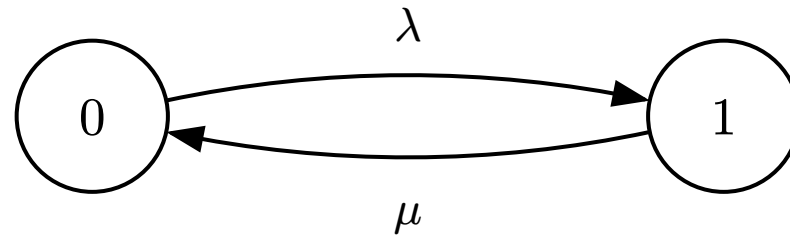
Theorem 1.5. *When the limiting distribution π exists, it is the unique solution to*

$$\begin{aligned}\lambda_0\pi_0 &= \mu_1\pi_1, \\ (\lambda_j + \mu_j)\pi_j &= \lambda_{j-1}\pi_{j-1} + \mu_{j+1}\pi_{j+1}, \quad j \geq 1, \\ \sum_{j=0}^{\infty} \pi_j &= 1.\end{aligned}$$

Remark: *This is quite intuitive, as “Rate in = Rate out” with “Rate” being the number of events per unit time.*

Problem

Consider a CTMC



with $\lambda, \mu > 0$. Derive the long-run proportion of time spent in state 0.

Theorem 1.6. *For a birth-and-death process without absorbing states, we have*

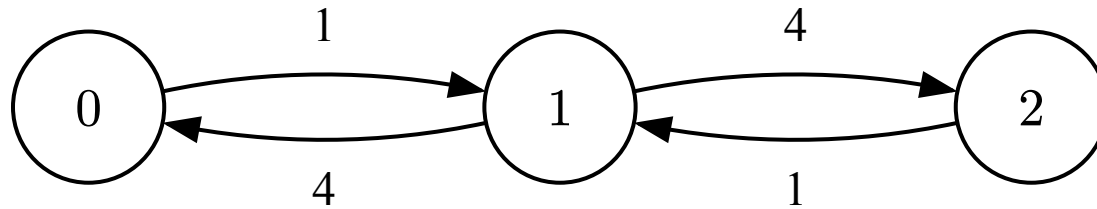
$$\pi_i = \theta_i \pi_0 = \frac{\theta_i}{\sum_{k=0}^{\infty} \theta_k}, \quad i = 0, 1, \dots,$$

where $\theta_0 = 1$ and

$$\theta_i = \frac{\lambda_0}{\mu_1} \cdot \frac{\lambda_1}{\mu_2} \cdot \dots \cdot \frac{\lambda_{i-1}}{\mu_i} = \frac{\prod_{j=0}^{i-1} \lambda_j}{\prod_{j=1}^i \mu_j}, \quad i = 1, 2, \dots$$

Problem

Consider a CTMC



Calculate π_0 , π_1 and π_2 .

Problem: Exam 2021 (Part 1)

Customers arrive in a store according to a Poisson process with rate $\lambda = 60$ customers per hour. The store has one server, and all customers in the service area are either in service or waiting in a queue. An arrival begins service immediately if the server is idle. If the server is busy, the customer joins the queue with probability 1 when there are at most two customers in the service area. With three or more, the customer joins the queue with probability $0 < p < 1$, or leaves immediately with probability $1 - p$. Joining decisions are independent across customers. Service times are i.i.d. exponential with mean $1/\mu = 6$ minutes and are independent of the arrival process. Whenever service is completed, the customer at the front of the queue immediately begins service.

Let $X(t)$ denote the number of customers either queuing or receiving service at time t .

- Assume that time is measured in hours, and determine the birth rates and the death rates as functions of p . Draw the transition diagram.
- Let $p = 0.1$, and assume that there are currently three customers queuing or receiving service.
 - Calculate the probability that exactly two customers finish their service times before the next customer joins the queue.
 - Calculate the expected time until the next customer joins the queue.

Theorem 1.7. *For a birth-and-death process $\{X(t) : t \geq 0\}$ with state space $\mathcal{S}$, let $A \subset \mathcal{S}$ and let*

$$v_i = \mathbb{E}[\min\{t \geq 0 : X(t) \in A\} \mid X(0) = i], \quad i \in \mathcal{S}.$$

If there are no absorbing states outside A , the expected times can be found by solving

$$\begin{aligned} v_i &= 0, & i \in A, \\ v_i &= \mathbb{E}[S_i] + \sum_{j \neq i} P(i \rightarrow j) v_j = \frac{1}{\lambda_i + \mu_i} + \sum_{j \neq i} P(i \rightarrow j) v_j, & i \notin A. \end{aligned}$$

Remark: *This is close to the first-step analysis we did for DTMCs.*

Problem: Exam 2021 (Part 2)

Customers arrive in a store according to a Poisson process with rate $\lambda = 60$ customers per hour. The store has one server, and all customers in the service area are either in service or waiting in a queue. An arrival begins service immediately if the server is idle. If the server is busy, the customer joins the queue with probability 1 when there are at most two customers in the service area. With three or more, the customer joins the queue with probability $0 < p < 1$, or leaves immediately with probability $1 - p$. Joining decisions are independent across customers. Service times are i.i.d. exponential with mean $1/\mu = 6$ minutes and are independent of the arrival process. Whenever service is completed, the customer at the front of the queue immediately begins service.

- Derive the limiting probabilities $\pi_i = \lim_{t \rightarrow \infty} P(X(t) = i)$ for $i = 0, 1, \dots$ as functions of p .
- What is the necessary condition on p for $\{X(t) : t \geq 0\}$ to have a limiting distribution? Give an intuitive explanation for the condition.

Connection between CTMCs and DTMCs

Theorem 1.8. *For a birth-and-death process with absorbing states, the absorption probabilities are computed using the discrete-time Markov chain with one-step transition probabilities given by*

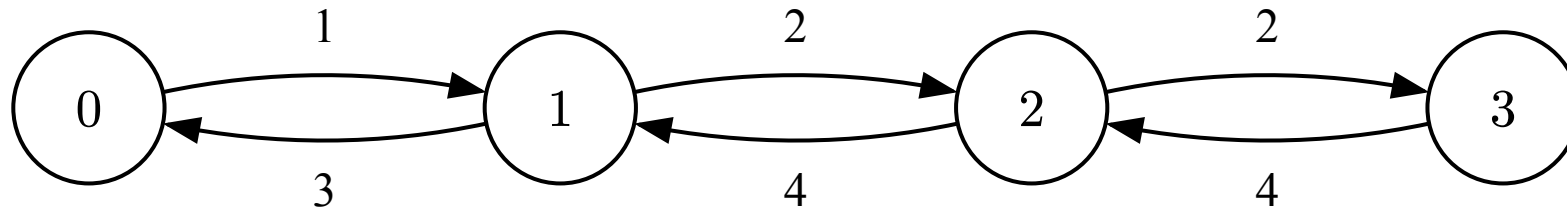
$$P_{i,j} = P(i \rightarrow j), \quad i \neq j,$$

and

$$P_{i,i} = \begin{cases} 0 & \text{if } \lambda_i + \mu_i > 0 \\ 1 & \text{if } \lambda_i + \mu_i = 0 \end{cases}, \quad i = 0, 1, \dots$$

Problem

Consider a CTMC



Let $X(0) = 1$, compute the probability that we reach state 3 before state 0.

Info about interactive lectures.

- I am going on a research visit to Emory University in Atlanta.
- No interactive lecture from week 41 to week 45.
- I will be back for week 46, and continue interactive lectures then.
- If any questions, ask on forum or by email.
- Good luck!